



User Manual

Version 0.2.0

*Provenance-aware reconciliation of taxonomic concepts
and molecular sequence records*

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Software documentation for TaxonWeave v0.2.0

Document information

Item	Details
Software	TaxonWeave
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Author	Andrew Davinack
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Core external data services	World Register of Marine Species (WoRMS) and GenBank
Core Python dependencies	Biopython >=1.81; pandas >=2.0; requests >=2.31
License	MIT License (as stated in the project README)
Manual status	User manual for the v0.2.0 implementation supplied with this documentation set

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1. Introduction and scope

TaxonWeave is a Python framework for provenance-aware reconciliation of taxonomic concepts and molecular sequence records. Its core problem is straightforward but consequential: taxonomic names change through time, whereas molecular records in public databases preserve names and annotations associated with database records. Sequence evidence relevant to a currently accepted marine taxon can therefore be distributed across the accepted name, historical synonyms, and other nomenclatural combinations.

TaxonWeave connects taxonomic information from the World Register of Marine Species (WoRMS) with nucleotide records from GenBank. It resolves a submitted scientific name to a current WoRMS concept, obtains the associated nomenclatural synonyms, searches GenBank across those names, deduplicates overlapping retrievals, preserves the organism's name represented on each GenBank record, and reconciles that deposited name against the current WoRMS concept.

The framework is designed as an evidence-reconciliation layer. It does not replace taxonomic judgment, infer species boundaries, validate sequence identifications, or perform a taxonomic revision. Instead, it makes relationships among taxonomic names, search pathways, molecular records, and associated metadata explicit and reproducible.

1.1 Core capabilities

- Resolve a scientific name through WoRMS and identify the currently accepted concept and AphiaID.
- Retrieve the WoRMS classification and nomenclatural synonyms associated with that concept.
- Search GenBank using the accepted name and the WoRMS synonym set.
- Record which taxonomic search name or names retrieved each GenBank record.
- Deduplicate GenBank records retrieved through multiple taxonomic names.
- Preserve the organism name represented on the GenBank record.
- Reconcile deposited names against the accepted name and WoRMS synonym set.
- Extract molecular, geographic, collection, specimen, and publication metadata.
- Conservatively normalize common molecular-marker annotations while retaining raw annotations.
- Export a five-file CSV report for a single taxon.
- Run the same reconciliation engine across multiple taxa and produce a comparative batch summary.

1.2 Current scope

The core v0.2.0 workflow is WoRMS -> taxonomic concept and nomenclatural history -> GenBank molecular records and metadata. Although a gbif.py module is present in the supplied source tree, GBIF occurrence-data integration is not part of the core query, CLI, report, or batch workflow documented here.

Note. *TaxonWeave relies on live external services. Results can change when WoRMS or GenBank records are added, revised, reannotated, or become temporarily unavailable.*

2. Conceptual model

The central design principle is separation of taxonomic concept, search provenance, and original database assertion. TaxonWeave does not treat the name used to discover a GenBank record as proof that the record was deposited under that name.

Concept	Meaning in TaxonWeave
query_name	The scientific name supplied by the user.
accepted_name	The currently accepted scientific name returned after WoRMS resolution.

found_via_name	The taxonomic search name or names through which a GenBank record was retrieved.
deposited_name	The organism name represented on the GenBank record itself.
name_relationship	TaxonWeave classification of the deposited name relative to the current WoRMS concept.
reconciliation_status	Higher-level state describing whether the deposited name is reconciled, nomenclaturally reconciled, or unresolved.

The workflow can be represented as:

```

User-supplied scientific name
|
v
WoRMS resolution
|-- matched record
|-- accepted concept / AphialD
|-- classification
`-- synonyms
|
v
GenBank searches: accepted name + synonyms
|
v
Deduplicate records while preserving search provenance
|
v
Parse deposited name, markers, geography, voucher/sample,
collection and publication metadata
|
v
Reconcile deposited name against accepted name + WoRMS synonyms
|
v
TaxonWeaveReport -> summary and CSV exports

```

2.1 Why search provenance matters

A historical synonym can retrieve a GenBank record even when the record itself is represented under the currently accepted name. TaxonWeave therefore preserves `found_via_name` separately from `deposited_name`. This prevents synonym-mediated discovery from being rewritten as an original taxonomic assertion.

2.2 Nomenclature is not independent biological validation

When a deposited name exactly matches a WoRMS synonym, TaxonWeave records a nomenclatural reconciliation. This means WoRMS currently asserts the synonym relationship. It does not independently establish that a particular sampled organism or sequence is biologically conspecific with the accepted concept.

3. Installation and verification

3.1 Requirements

TaxonWeave v0.2.0 requires Python 3.9 or later. Package dependencies declared in `pyproject.toml` are Biopython `>=1.81`, pandas `>=2.0`, and requests `>=2.31`.

3.2 Install from PyPI

```
python -m pip install taxonweave
```

Verify the installation:

```
python -c "import taxonweave; print(taxonweave.__version__)"
taxonweave --help
```

For the documented release, the version command should report 0.2.0.

3.3 Development installation

For source development, clone the repository, create a virtual environment, and install in editable mode:

```
git clone https://github.com/parasiteguy/taxonweave.git
cd taxonweave
python -m venv .venv
source .venv/bin/activate
python -m pip install -e .
```

Note. On Windows, virtual-environment activation uses the platform-specific activation script rather than source .venv/bin/activate.

4. Quick start

4.1 Single taxon from the command line

```
taxonweave query "Ficopomatus enigmaticus"
```

The command resolves the supplied name through WoRMS, constructs the accepted-name-plus-synonyms GenBank search, retrieves unique nucleotide records, reconciles deposited names, and prints a human-readable summary.

4.2 Export the report

```
taxonweave query "Ficopomatus enigmaticus" --export
```

To choose the output directory:

```
taxonweave query "Ficopomatus enigmaticus" \
--export \
--output ficopomatus_results
```

4.3 Query from Python

```
from taxonweave import query_species

report = query_species("Ficopomatus enigmaticus")
report.summary()
report.export("ficopomatus_results")
```

5. Command-line interface

The console entry point is taxonweave. Version 0.2.0 exposes two subcommands: query and batch.

5.1 query

```
taxonweave query SCIENTIFIC_NAME [--export] [--output OUTPUT]
```

Argument	Required	Meaning
----------	----------	---------

scientific_name	Yes	Scientific name supplied to the WoRMS-first reconciliation workflow.
--export	No	Export the report as CSV files.
--output OUTPUT	No	Output directory used only when --export is supplied.

If --export is used without --output, TaxonWeave constructs a directory from the accepted name by replacing spaces with underscores and appending _report.

5.2 batch

```
taxonweave batch INPUT_FILE [--output OUTPUT]
```

Argument	Required	Meaning
input_file	Yes	CSV file containing a column named exactly scientific_name.
--output OUTPUT	No	Directory for batch results. Default: taxonweave_batch.

6. Python API and the TaxonWeaveReport object

The public package API exports query_species. Calling query_species(scientific_name) returns a TaxonWeaveReport after the complete WoRMS-GenBank workflow has run.

```
from taxonweave import query_species

report = query_species("Alitta succinea")
```

6.1 Report attributes

Attribute	Contents
query_name	Original name supplied to query_species().
query_record	WoRMS record matched to the supplied name.
accepted_record	Resolved current WoRMS record.
accepted_name	scientificname from the accepted WoRMS record.
aphia_id	AphiaID from the accepted WoRMS record.
taxonomy	Flattened WoRMS classification as a list of dictionaries.
synonyms	WoRMS synonym records associated with the accepted AphiaID.
search_results	Mapping of each GenBank search name to retrieved identifiers.
genbank	pandas DataFrame containing parsed and reconciled GenBank records.

6.2 Report methods

Method	Purpose
summary_dict()	Return the standard summary statistics as a Python dictionary.

<code>summary()</code>	Print a human-readable reconciliation report.
<code>export(directory_name=None)</code>	Write <code>taxonomy.csv</code> , <code>synonyms.csv</code> , <code>genbank_searches.csv</code> , <code>sequences.csv</code> , and <code>summary.csv</code> .

7. WoRMS taxonomic resolution

TaxonWeave begins with WoRMS rather than searching GenBank directly. The supplied name is queried using the WoRMS REST service with `exact-name` matching (`like=false`) and `marine_only=false`. If no record is returned, `query_species` raises a `ValueError`.

7.1 Accepted versus unaccepted input names

If the matched WoRMS record has status `accepted`, it becomes the current concept directly. Otherwise, TaxonWeave uses `valid_AphiaID`, when available, to retrieve the accepted record. This permits an obsolete or historical name to serve as the user input while still anchoring downstream searches to the current accepted concept.

7.2 Classification

WoRMS classification is retrieved by `AphiaID` and flattened from the nested response into rows containing rank, `scientificname`, and `AphiaID`. These rows become `taxonomy.csv`.

7.3 Synonyms

All records returned by the WoRMS synonym endpoint for the accepted `AphiaID` are retained in the report. For GenBank searching, TaxonWeave extracts unique nonblank `scientificname` values from those synonym records and combines them with the accepted name.

8. GenBank retrieval and search provenance

TaxonWeave searches GenBank nucleotide records separately for the accepted name and each unique WoRMS synonym. Search results are then combined into a unique identifier set while retaining a mapping between each record and the taxonomic search name or names that retrieved it.

8.1 Deduplication

A record may be returned by more than one taxonomic search. TaxonWeave deduplicates the underlying GenBank identifiers before downloading records, but preserves all associated search names as provenance. For a record found through multiple names, `found_via_name` is represented as a semicolon-separated string in the parsed record.

8.2 Record parsing

Each downloaded GenBank record is parsed into a standardized evidence row. The parser obtains the organism name from the source feature's organism qualifier when available, with the record-level organism annotation as a fallback. The resulting `deposited_name` is kept separate from `accepted_name` and `found_via_name`.

8.3 Entrez configuration

The supplied GenBank module contains `configure_entrez(email, api_key=None)`, which assigns Biopython `Entrez.email` and, optionally, `Entrez.api_key`. Users embedding TaxonWeave in larger workflows should follow NCBI requirements and practices for Entrez identification and rate limits.

9. Taxonomic reconciliation

After GenBank records are downloaded and parsed, TaxonWeave compares each deposited_name with the accepted WoRMS name and the WoRMS synonym set. Comparison is deliberately conservative: names are stripped, converted to lowercase, and internal whitespace is normalized, but TaxonWeave does not perform additional taxonomic inference.

name_relationship	reconciliation_status	Interpretation
ACCEPTED_NAME	RECONCILED	Deposited name exactly matches the current accepted WoRMS scientific name after conservative string normalization.
WORMS_SYNONYM	RECONCILED_NOMENCLATURALLY	Deposited name exactly matches a name currently returned by WoRMS as a synonym of the accepted taxon.
UNRESOLVED_NAME	UNRESOLVED	A deposited organism name exists but does not exactly match the accepted name or supplied WoRMS synonym set.
MISSING_NAME	UNRESOLVED	No deposited organism name was available.

Note. *WORMS_SYNONYM* is a nomenclatural result. It is not independent evidence validating the biological identity of the sequence.

10. Molecular-marker interpretation

GenBank annotations are heterogeneous. TaxonWeave extracts raw marker-related annotations and applies conservative normalization rules. The raw evidence is retained in marker_raw while interpreted values are stored in marker_normalized, marker_category, and normalization_status.

Status	Meaning
NORMALIZED	Annotation supports a standardized marker designation.
RECOGNIZED	Annotation is biologically interpretable but is not forced into a traditional phylogenetic-marker abbreviation.
AMBIGUOUS	Annotation is meaningful but insufficiently specific for a defensible standardized assignment.
UNRESOLVED	Annotation cannot currently be interpreted reliably.

10.1 Examples of normalized markers

The implementation contains explicit normalization rules for commonly encountered markers including COI, COII, COIII, CytB, ATP6, ATP8, ND1-ND6, ND4L, 12S and 16S mitochondrial rRNA, 18S and 28S nuclear rRNA, and histone H3.

10.2 Conservative treatment of ambiguity

TaxonWeave intentionally avoids forcing underspecified annotations into a more specific marker identity. For example, the project documentation explicitly notes that a generic annotation such as 'large subunit ribosomal RNA' should remain ambiguous rather than automatically being interpreted as 28S.

11. Metadata extraction

In addition to taxonomic and marker information, each parsed GenBank record can contain the following evidence classes.

Evidence class	Fields
Sequence	accession, accession_base, sequence_length, molecule_type, topology
Taxonomic provenance	deposited_name, accepted_name, found_via_name, name_relationship, reconciliation_status
Molecular evidence	marker_raw, marker_normalized, marker_category, normalization_status
Geography and collection	geo_loc_name, country, locality, lat_lon, collection_date, collected_by, identified_by
Specimen/sample provenance	voucher_raw, voucher_present, isolate, strain, clone, isolation_source, source_note, db_xref
Publication provenance	reference_title, reference_journal, reference_authors

11.1 Voucher interpretation

`voucher_present` is True when a `specimen_voucher` qualifier is present. This indicates the presence of that GenBank annotation; it does not verify the existence, accessibility, correctness, or taxonomic identity of the cited voucher.

11.2 Geographic metadata

TaxonWeave preserves `geo_loc_name`, `country`, `locality`, and `lat_lon` as supplied by the record when available. The v0.2.0 core workflow does not geocode, standardize place names, or validate coordinates.

12. Summary statistics

`TaxonWeaveReport.summary_dict()` returns the following standard query-level metrics.

Metric	Definition
<code>query_name</code>	Original submitted scientific name.
<code>accepted_name</code>	Resolved accepted WoRMS scientific name.
<code>aphia_id</code>	Accepted WoRMS AphiaID.
<code>worms_synonyms</code>	Number of synonym records returned by WoRMS.
<code>genbank_search_names</code>	Number of taxonomic names searched in GenBank.
<code>search_names_with_hits</code>	Number of search names returning at least one identifier.
<code>genbank_records</code>	Number of parsed unique GenBank records.
<code>deposited_taxon_names</code>	Number of unique nonmissing <code>deposited_name</code> values.
<code>accepted_name_records</code>	Records classified <code>ACCEPTED_NAME</code> .
<code>worms_synonym_records</code>	Records classified <code>WORMS_SYNONYM</code> .
<code>unresolved_name_records</code>	Records classified <code>UNRESOLVED_NAME</code> .
<code>missing_name_records</code>	Records classified <code>MISSING_NAME</code> .

marker_normalized_records	Occurrences of NORMALIZED in per-record marker status.
marker_recognized_records	Occurrences of RECOGNIZED in per-record marker status.
marker_ambiguous_records	Occurrences of AMBIGUOUS in per-record marker status.
marker_unresolved_records	Occurrences of UNRESOLVED in per-record marker status.
geographic_records	Records with nonmissing geo_loc_name.
coordinate_records	Records with nonmissing lat_lon.
collection_date_records	Records with nonmissing collection_date.
voucher_records	Records with voucher_present evaluating to True.

Note. Marker-status counts are accumulated from semicolon-separated `normalization_status` values. A record containing multiple interpreted marker annotations can therefore contribute to more than one status occurrence; these counts should not automatically be treated as mutually exclusive record categories.

13. Exported report files

A single-taxon export contains five CSV files.

File	Purpose
taxonomy.csv	Flattened WoRMS classification for the accepted concept.
synonyms.csv	WoRMS synonym records associated with the accepted AphiaID.
genbank_searches.csv	One row per taxonomic GenBank search name, with retrieval count and hit/no-hit indicator.
sequences.csv	Record-level molecular evidence, metadata, search provenance, and taxonomic reconciliation.
summary.csv	One-row summary generated by <code>summary_dict()</code> .

13.1 genbank_searches.csv

Column	Meaning
search_name	Accepted name or WoRMS synonym used in the GenBank search.
records_retrieved	Number of identifiers returned for that search name.
retrieved_records	Boolean indicating whether the search returned any identifiers.

13.2 sequences.csv

`sequences.csv` is the principal record-level output. It deliberately combines source-derived fields with TaxonWeave-derived fields so downstream users can distinguish original database information from reconciliation and normalization results. A complete column dictionary is provided in Appendix C.

14. Batch querying

The batch layer is an orchestration wrapper around `query_species()`; it does not implement a second reconciliation algorithm. Each input name is processed sequentially using the same single-taxon engine, and each successful query is exported to its own directory.

14.1 Input file

```
scientific_name
Alitta succinea
Ficopomatus enigmaticus
Boccardia proboscidea
```

The column name `scientific_name` is required exactly. Blank values are skipped. Duplicate nonblank names are retained because each row is treated as a submitted query.

14.2 Output structure

```
results/
|-- batch_summary.csv
|-- Alitta_succinea/
|   |-- taxonomy.csv
|   |-- synonyms.csv
|   |-- genbank_searches.csv
|   |-- sequences.csv
|   |-- summary.csv
|-- Ficopomatus_enigmaticus/
|   |-- ...
|-- Boccardia_proboscidea/
|   |-- ...
```

14.3 Duplicate output concepts

Directory names are generated from the accepted name (or submitted name as fallback) using a filesystem-safe transformation. If the target directory already exists, TaxonWeave appends `_2`, `_3`, and so forth rather than overwriting the previous result.

14.4 Comparative metrics

Field	Calculation
accepted_name_pct	<code>accepted_name_records / genbank_records x 100</code>
synonym_name_pct	<code>worms_synonym_records / genbank_records x 100</code>
unresolved_name_pct	<code>unresolved_name_records / genbank_records x 100</code>
geographic_metadata_pct	<code>geographic_records / genbank_records x 100</code>
coordinate_metadata_pct	<code>coordinate_records / genbank_records x 100</code>
collection_date_pct	<code>collection_date_records / genbank_records x 100</code>
voucher_metadata_pct	<code>voucher_records / genbank_records x 100</code>

Percentages are rounded to one decimal place. When `genbank_records` is zero, the percentage function returns 0.0.

14.5 Failure handling

A failure for one submitted taxon does not terminate the batch. The failed row is written with `batch_status=FAILED`, a `batch_error` containing the exception type and message, and no output directory; processing continues with subsequent names. Successful rows receive `batch_status=SUCCESS` and the corresponding output-directory name.

15. Worked examples

15.1 Accepted-name query: *Ficopomatus enigmaticus*

```
taxonweave query "Ficopomatus enigmaticus" --export --output ficopomatus_results
```

The project documentation uses *Ficopomatus enigmaticus* as a representative case because WoRMS recognizes historical names including *Mercierella enigmatica* and *Phycopomatus enigmaticus*. TaxonWeave searches the accepted name and returned synonyms independently, records which names retrieve each sequence, deduplicates overlapping records, and preserves each record's deposited taxonomic name.

15.2 Historical-name query

```
taxonweave query "Mercierella enigmatica"
```

The submitted historical name is first resolved through WoRMS. The remainder of the workflow is then constructed from the accepted concept and its WoRMS synonym set, rather than treating the submitted historical name as the final taxonomic authority.

15.3 Batch audit

```
taxonweave batch species.csv --output results
```

The combined `batch_summary.csv` can be used to compare molecular-record representation and metadata completeness across taxa. These percentages describe the records retrieved by the TaxonWeave workflow at the time of the query; they are not estimates of biological sampling completeness.

16. Scientific interpretation

16.1 What TaxonWeave supports

- Auditing how molecular records associated with a current taxonomic concept are distributed across nomenclatural search pathways.
- Separating the name that discovered a record from the name represented on the record.
- Identifying records whose deposited names are directly reconciled, nomenclaturally reconciled, unresolved, or missing relative to a current WoRMS concept.
- Auditing presence of common GenBank metadata fields such as coordinates, collection dates, and specimen_voucher qualifiers.
- Organizing molecular-marker annotations into conservative normalized, recognized, ambiguous, and unresolved states.
- Producing reproducible tabular outputs suitable for downstream review and analysis.

16.2 What TaxonWeave does not establish

- It does not delimit species or infer cryptic lineages.

- It does not determine whether a GenBank identification is biologically correct.
- It does not independently validate a WoRMS synonymy.
- It does not verify museum vouchers merely because a specimen_voucher qualifier is present.
- It does not infer absence of a marker, voucher, locality, or sampling event outside the records retrieved by the workflow.
- It does not make the live external databases static; rerunning a query later can produce different results.

17. Reproducibility and provenance

TaxonWeave's reproducibility model is based on preserving database assertions and derived interpretations separately. Raw taxonomic and molecular annotations are retained where the implementation exposes them, while derived fields identify search provenance, marker interpretation, and taxonomic reconciliation.

17.1 Version recording

For analyses intended for publication, record the TaxonWeave version, query date, input names, and exported report directory. Because WoRMS and GenBank are live services, the software version alone is insufficient to guarantee identical future retrieval results.

17.2 Recommended archival practice

For a manuscript or reproducible analysis, retain the exported CSV report(s) used in the analysis alongside scripts or notebooks that consume them. This preserves the evidence snapshot even if the upstream databases later change.

18. Error handling and troubleshooting

Problem	Likely cause / behavior	Action
WoRMS could not resolve a name	resolve_name returned no records.	Check spelling and nomenclature; try a valid scientific name represented in WoRMS.
Accepted concept could not be resolved	No usable accepted record was obtained.	Inspect the WoRMS record and valid_AphiaID relationship.
No GenBank records	All taxonomic search names returned no identifiers.	This is a valid zero-hit outcome; inspect genbank_searches.csv if exporting.
Network/API error	WoRMS or NCBI request failed or timed out.	Retry later and confirm internet/API availability.
Batch CSV rejected	File missing, lacks scientific_name, or contains no nonblank names.	Correct the input path/column/content.
One batch taxon fails	An exception occurred for that submitted row.	Inspect batch_error; other taxa continue processing.
Unexpected unresolved deposited names	Deposited name does not exactly match accepted name or WoRMS synonym set after conservative normalization.	Review the record manually; do not assume unresolved means incorrect.

18.1 Command help

```
taxonweave --help
taxonweave query --help
taxonweave batch --help
```

19. Known limitations

- Taxonomic reconciliation uses exact normalized string comparison against the accepted WoRMS name and returned synonym names; it does not perform fuzzy matching or broader taxonomic inference.
- External database content is mutable. Counts and metadata can change between runs.
- Presence of a GenBank qualifier is not equivalent to independent validation of that metadata.
- Marker normalization is rule-based and intentionally conservative; uncommon or underspecified annotations may remain ambiguous or unresolved.
- The core v0.2.0 workflow does not integrate occurrence records from GBIF, despite the presence of a separate gbif.py module in the supplied source.
- Batch processing catches per-taxon exceptions to preserve continuation. Users should inspect batch_status and batch_error rather than assuming every submitted row succeeded.
- The framework does not perform automated species delimitation, phylogenetic inference, or taxonomic revision.

20. Development and testing

The supplied project includes tests for package integrity, reconciliation, marker interpretation, CLI parsing, and batch behavior.

```
pytest -v
```

Deterministic unit tests should be distinguished from live database queries. External API content can change independently of TaxonWeave and should therefore not be treated as a stable unit-test fixture.

20.1 Source modules

Module	Role
query.py	Integrated WoRMS-GenBank workflow, reconciliation, report object, summaries, exports.
worms.py	WoRMS REST resolution, accepted-record retrieval, classification, and synonyms.
genbank.py	Entrez searching, deduplication support, record fetching/parsing, marker interpretation, metadata extraction.
batch.py	Multi-taxon orchestration and comparative summary metrics.
cli.py	argparse-based query and batch command-line interface.
__init__.py	Public query_species import and package version.
gbif.py	Separate GBIF-related code present in source but not part of the core v0.2.0 workflow.

21. Citation, versioning, and license

21.1 Version

This manual documents TaxonWeave v0.2.0.

21.2 Citation

A formal software-paper citation was not supplied in the source materials used to prepare this manual. Until a formal citation is established, users should cite the software version and repository/release information appropriate to the distributed project.

21.3 License

The project README states that TaxonWeave is released under the MIT License.

Appendix A. Command-line reference

```
taxonweave [-h] {query,batch} ...
```

```
taxonweave query [-h] [--export] [--output OUTPUT] scientific_name
```

```
taxonweave batch [-h] [--output OUTPUT] input_file
```

Command	Example
Single query	taxonweave query "Alitta succinea"
Single query + export	taxonweave query "Alitta succinea" --export
Custom export directory	taxonweave query "Alitta succinea" --export --output alitta_results
Batch	taxonweave batch species.csv --output results

Appendix B. Status terminology

Field	Value	Meaning
name_relationship	ACCEPTED_NAME	Deposited name matches accepted WoRMS name.
name_relationship	WORMS_SYNONYM	Deposited name matches a WoRMS synonym.
name_relationship	UNRESOLVED_NAME	Deposited name is present but matches neither accepted name nor synonym set.
name_relationship	MISSING_NAME	No deposited organism name available.
reconciliation_status	RECONCILED	Direct accepted-name match.
reconciliation_status	RECONCILED_NOMENCLATURALLY	Match mediated by current WoRMS synonymy.
reconciliation_status	UNRESOLVED	No accepted/synonym reconciliation established.
normalization_status	NORMALIZED	Standardized marker designation supported.
normalization_status	RECOGNIZED	Biologically interpretable without forcing a standard marker abbreviation.
normalization_status	AMBIGUOUS	Meaningful but insufficiently specific for standardized assignment.
normalization_status	UNRESOLVED	Cannot currently be interpreted reliably.
batch_status	SUCCESS	Taxon query completed and was exported.
batch_status	FAILED	Taxon query raised an exception; details stored in batch_error.

Appendix C. sequences.csv data dictionary

Column	Definition	Provenance
--------	------------	------------

accession	GenBank accession, including version when present.	Source-derived
accession_base	Accession without the version suffix.	TaxonWeave-derived
sequence_length	Length of the downloaded sequence.	Source-derived/computed from record
molecule_type	GenBank molecule_type annotation.	Source-derived
topology	GenBank topology annotation.	Source-derived
deposited_name	Organism name represented on the GenBank record.	Source-derived
accepted_name	Current accepted name resolved through WoRMS.	WoRMS-derived
found_via_name	Taxonomic search name(s) that retrieved the record.	TaxonWeave provenance
marker_raw	Raw marker-related annotation(s) interpreted from the record.	Source-derived aggregation
marker_normalized	Conservatively standardized marker designation(s), when supported.	TaxonWeave-derived
marker_category	Biological category assigned during marker interpretation.	TaxonWeave-derived
normalization_status	Marker interpretation status(es).	TaxonWeave-derived
geo_loc_name	GenBank geo_loc_name qualifier.	Source-derived
country	GenBank country qualifier, when present.	Source-derived
locality	GenBank locality qualifier, when present.	Source-derived
lat_lon	GenBank lat_lon qualifier.	Source-derived
collection_date	GenBank collection_date qualifier.	Source-derived
collected_by	GenBank collected_by qualifier.	Source-derived
identified_by	GenBank identified_by qualifier.	Source-derived
voucher_raw	GenBank specimen_voucher qualifier.	Source-derived
voucher_present	Boolean indicating whether specimen_voucher is nonblank.	TaxonWeave-derived
isolate	GenBank isolate qualifier.	Source-derived
strain	GenBank strain qualifier.	Source-derived
clone	GenBank clone qualifier.	Source-derived
isolation_source	GenBank isolation_source qualifier.	Source-derived
source_note	Joined source-feature note qualifier(s).	Source-derived aggregation
db_xref	Joined source-feature db_xref qualifier(s).	Source-derived aggregation
reference_title	Reference title extracted from the GenBank record.	Source-derived
reference_journal	Reference journal field extracted from the GenBank record.	Source-derived
reference_authors	Reference author field extracted from the GenBank record.	Source-derived

name_relationship	Deposited-name relationship to current WoRMS concept.	TaxonWeave-derived
reconciliation_status	Higher-level reconciliation state.	TaxonWeave-derived

Appendix D. summary.csv and batch_summary.csv data dictionary

summary.csv column	Definition
query_name	Original submitted scientific name.
accepted_name	Resolved accepted WoRMS scientific name.
aphia_id	Accepted WoRMS AphiaID.
worms_synonyms	Number of synonym records returned by WoRMS.
genbank_search_names	Number of taxonomic names searched in GenBank.
search_names_with_hits	Number of search names returning at least one identifier.
genbank_records	Number of parsed unique GenBank records.
deposited_taxon_names	Number of unique nonmissing deposited_name values.
accepted_name_records	Records classified ACCEPTED_NAME.
worms_synonym_records	Records classified WORMS_SYNONYM.
unresolved_name_records	Records classified UNRESOLVED_NAME.
missing_name_records	Records classified MISSING_NAME.
marker_normalized_records	Occurrences of NORMALIZED in per-record marker status.
marker_recognized_records	Occurrences of RECOGNIZED in per-record marker status.
marker_ambiguous_records	Occurrences of AMBIGUOUS in per-record marker status.
marker_unresolved_records	Occurrences of UNRESOLVED in per-record marker status.
geographic_records	Records with nonmissing geo_loc_name.
coordinate_records	Records with nonmissing lat_lon.
collection_date_records	Records with nonmissing collection_date.
voucher_records	Records with voucher_present evaluating to True.

batch_summary.csv contains the standard summary fields for successful queries plus the following batch-specific fields:

Batch field	Definition
accepted_name_pct	Percent of unique GenBank records classified ACCEPTED_NAME.
synonym_name_pct	Percent classified WORMS_SYNONYM.
unresolved_name_pct	Percent classified UNRESOLVED_NAME.
geographic_metadata_pct	Percent with nonmissing geo_loc_name.

coordinate_metadata_pct	Percent with nonmissing lat_lon.
collection_date_pct	Percent with nonmissing collection_date.
voucher_metadata_pct	Percent with voucher_present=True.
batch_status	SUCCESS or FAILED.
batch_error	Exception type and message for failed queries; otherwise empty/None.
output_directory	Per-taxon report directory name for successful queries.